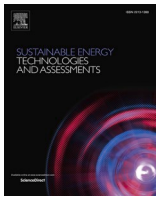
Sustainable Energy Technologies and Assessments 54 (2022) 102823



Contents lists available at ScienceDirect

Sustainable Energy Technologies and Assessments

journal homepage: www.elsevier.com/locate/seta





Stabilization of frequency in Multi-Microgrid system using barnacle mating Optimizer-based cascade controllers

K. Peddakapu [a,*], P. Srinivasarao [b], M.R. Mohamed [a,c], Yogendra Arya [d], D.J. Krishna Kishore [a,c]

[a] Centre for Advanced Industrial Technology, Universiti Malaysia Pahang, 26600 Pekan, Malaysia
[b] Department of Electrical & Electronics Eng., Nalanda Institute of Eng. & Tech., AP, India
[c] Faculty of Electrical and Electronics Eng. Tech., Universiti Malaysia Pahang, 26600 Pekan, Malaysia
[d] Department of Electrical Engineering, J.C. Bose University of Science and Technology, YMCA, Haryana, India

ARTICLE INFO

Keywords:
Automatic generation control
Microgrid
Redox flow battery
Renewable energy sources
Barnacle mating optimizer

ABSTRACT

This paper presents a new control strategy in multi-microgrid system for supressing the variations of frequency and of tie-line power flow due to intermittent nature of renewable sources and frequently varying load demands. A novel cascade combination of fractional order-based integral derivative and proportional integral derivative with filter called CFOID-FOPIDN controller is utilized as secondary controller for automatic generation control. In this work, diverse sources such as photovoltaic, wind, diesel engine, fuel-cell, electrical vehicles and battery energy storage systems are considered for investigations under various uncertainties conditions. A newly developed barnacle mating optimizer is presented to tune the CFOID-FOPIDN for diminishing the frequency and tie-line power changes. Moreover, the amalgamation of redox flow batteries and modern flexible alternating current transmission system device like distributed power flow controller is integrated into the system for ameliorating the dynamic stability of the system. The tested outcomes show that the barnacle mating optimizer-based CFOID-FOPIDN controller with the combination of distributed power flow controller and redox flow batteries exhibits low fluctuation rate for frequency (5.6 s for area-1 and 5.5 s for area 2) and tie-line power (6.2 s). Finally, the robustness of the CFOID-FOPIDN is validated under three operating conditions, such as step load perturbations, change in wind speed, irradiance, and change in both wind and irradiance levels.

Introduction

In the recent years, an uninterrupted increase of load demand, liberalization in the energy market, increasing of expenses for transmission and utilization, and deteriorating environmental circumstances have substantially accelerated the breakthrough of microgrids. Various renewable and green energy (RGE)-based power generating units and groups of loads at different locations can be defined as microgrids (MGs). The MG is an attainable alternative to endorse the incorporation of renewable and sustainable energy sources with conventional sources in order to reduce the dependency on traditional power sources [1,2]. Furthermore, the MG can be directed a significant role in upcoming interconnected power systems.

The privileges of the MGs are to provide a good quality of power without any violation in electrical parameters, to demonstrate the finer performance at all considerable loading conditions, to be required less cost for sustaining, and to be upgraded as quickly. Considering these advantages, exceptional communication and advanced controlling approaches have been transmuted the structure of the traditional isolated microgrid into a multi-microgrid (MMG) system with the assists of tie-lines [3,4]. The difficult objective of the MMGs is to be equalized the delivering power and utilizing load demand without affecting the frequency and voltage due to fast changes in atmospheric circumstances.

Moreover, the stability and controllability of the interconnected multi-area microgrid or multi-microgrid system is mainly depending on the scheduled values of frequency and voltage. Therefore, in the recent years, there has been an increasing interest in interconnected MMGs with automatic generation control (AGC) for mitigating the frequency variations [5,6]. The preliminary regulating activities of AGC are investigated by governor speed, but these are not sufficient to acceptably alleviating the oscillations against unpredicted perturbations. Regarding this, a complementary controller is required to subside the frequency and tie-line power changes, and hence to maintain the system dynamic stability.

The past ten years have witnessed increasingly rapid advances in the

* Corresponding author.
E-mail address: peddaump@gmail.com (K. Peddakapu).

https://doi.org/10.1016/j.seta.2022.102823
Received 7 March 2022; Received in revised form 10 September 2022; Accepted 28 September 2022
Available online 9 October 2022
2213-1388/© 2022 Elsevier Ltd. All rights reserved.

Nomenclature

MGs	Microgrids
AGC	Automatic generation control
RFBs	Redox flow batteries
FC	Fuel Cell
BESS	Battery energy storage system
DPFC	Distributed power flow controller
PID	Proportional integral derivative
2DOF	Two degree of freedom
FOC	Fractional order controller
SMC	Sliding mode controller
FLC	Fuzzy logic controller
PSO	Particle swarm optimization
BFOA	Bacterial foraging optimization algorithm
CS	Cuckoo search
DEG	Diesel engine generator
ISE	Integral of square error
US	Undershoot
ITSE	Integral time square error
MMG	Multi-microgrid
FACTS	Flexible AC transmission system
PV	Photovoltaic
EVs	Electric vehicles
PI	Proportional integral
ID	Integral derivative
IDD	Integral double derivative
FO	Fractional order
FOI	Fractional order integral
VSC	Variable structure controller
GA	Genetic algorithm
SSA	Salp swarm algorithm
BMO	Barnacle mating optimizer
WTG	Wind turbine generator
ITAE	Integral time absolute error
ST	Setting time
IAE	Integral absolute error
OS	Overshoot

field of multi-microgrid by several researchers for smoothening the fluctuations in tie-line and frequency. Traditional controllers like proportional integral (PI), integral double derivative (IDD), and proportional integral derivative (PID) are utilized in power industry due to their consistency and ease of implementation [7]. However, these traditional controllers can't be worked correctly when the microgrid connects with other microgrids and other changeable loading circumstances. To determine these difficulties, finer dynamic performance might have predicted from other control approaches, such as intelligent methods [8], decentralized controllers [9], optimal controllers [10], sliding mode controllers (SMC) [11], internal mode control [12], fuzzy logic controllers [13], and extended distributed model predictive control (EDMPC) [14] are used to abating the frequency variations. Nevertheless, many of these approaches require evidence of the state of the system, thus, they might be difficult to execute in real-time applications.

On the other hand, fractional-order (FO) type controllers have attracted much attention in recent years due to an adaptable structure and having more tuning parameters [15,16]. When the tuning parameters are increased then involuntary hyper-damped poles are improved. It denotes that the region of stability is enriched gradually, then has more adjustability to design the controller. Furthermore, FO-based controllers have used enormously in the interconnected power systems over recent years for dealing the AGC issues. Some researchers [17,18] have suggested the FOI, FOPID, FO-based fuzzy PID (FOFPID), FO-based FPID with filter (FOFPIDN), FO-based hybrid fuzzy logic controllers in the AGC of MMGs and power systems for enhancing the dynamic performance. Singh et al. [19] have proposed the dual stage FO-based PID controller for stabilizing changes in frequency of standalone microgrid system.

Conversely, few researchers [20,21] have implemented the cascaded-based FO controllers such as CF-FOIDN, FOPI-FOPID, FFOPI-FOPD, and FFOPI-FOPID have been presented in the AGC systems with incorporation of renewable and distributed energy sources. Moreover, 3^0 of freedom-based tilt integral derivative (3DOF-TID) and FOPID is cascaded for enhancing the frequency stability in microgrid system [22]. Gate controlled series capacitor (GCSC) is presented in multi-area power system with cascade connection of TID and FOPID controller [23]. A careful study of the literature reveals that the combination of FO-based cascade controllers has not been thoroughly investigated. Hence, it is required to be examined the AGC with a cascaded FO controller, and this is the motivation of the present study. In this work, a new cascade connection of FOID and FOPIDN (CFOID-FOPIDN) is used as secondary controller in the AGC of MMGs for dwindling the changes in frequency and tie-line power flows.

Requisite optimization methods are needed for ascertaining the optimal controller gains to stabilize the uncertainties in the system. Latif et al. [24,25] have presented the mine blast and sine–cosine algorithms for investigating the hybrid microgrid system with the PI-based one plus double derivative {PI-(1 + DD)} and IPD-based one plus integral {IPD-(1 + I)} controllers. Moreover, various optimizations such as genetic algorithm (GA)-based FOPID [26], particle swarm optimization (PSO) and bacterial foraging optimization algorithm (BFOA)-based PI/PID [27,28] are proposed for attaining the better dynamic performance of the systems. However, the GA, PSO, ICA and BFOA methods exhibit the slow convergence speed. Price-based demand response (PBDR) technique is employed in hybrid microgrid system with salp swarm algorithm (SSA) [29] for frequency control. Imperialist competitive algorithm (ICA) is designed to optimize the integral tilt derivative with filter (ITDF) in multi-microgrid system [30].

Another newly designed bio-nature inspired algorithm called barnacle mating optimizer (BMO) [31] is used to solve the difficulties of interconnected power systems in the case of uncertainties. Though these investigations disclosed many significant results, little work has been carried out with bio-nature inspired algorithms-based cascaded controllers in MMGs. Hence, newly developed BMO is adopted in this work to tune the suggested CFOID-FOPIDN controller for improving the system dynamic performance. Nevertheless, it is difficult to obtain performance enhancement in mitigating the abnormalities of frequency and tie-line power with only BMO tuned CFOID-FOPIDN due to the incorporation of various renewable, distributed energy sources and batteries in interconnected MMGs. Hence, another motivation for this study is to integrate the combination of advanced FACT device like distributed power flow controller (DPFC) and redox flow batteries (RFB) for dwindling the changes in tie-line power and frequency of MMGs.

The objectives of this work are specified below:

1. To investigate the AGC of MMG system with the incorporation of diesel engine, fuel cell (FC), photovoltaic (PV), electric vehicle (EV), battery energy storage system (BESS) and wind turbine generator (WTG).
2. To design a new BMO-based CFOID-FOPIDN controller for AGC of MMG system under study.
3. To validate the superiority of CFOID-FOPIDN over other control structures.

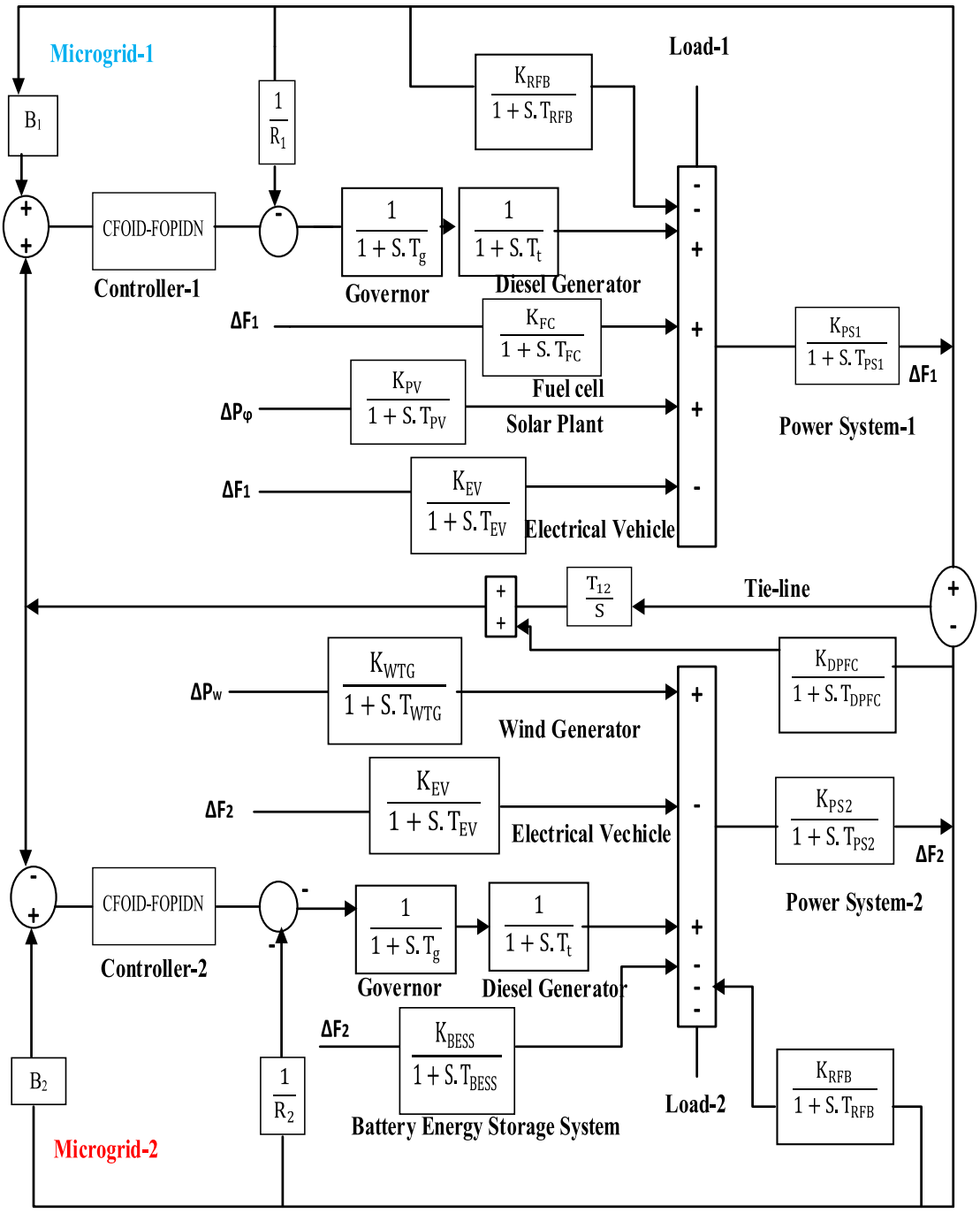


Fig. 1. Proposed interconnected multi-microgrid system.

4. To show the preeminence of BMO-based CFOID-FOPIDN over other optimization techniques-based CFOID-FOPIDN controller.
5. To inspect the positive impact of incorporation of DPFC and RFB for alleviating variations in frequency and tie-line power of MMG.
6. To verify the effectiveness of the proposed CFOID-FOPIDN controller under four conditions such as SLP, wind speed variations, irradiance variations, and variations in both wind speed and irradiance levels.

Contribution

The novelty and contributions of this work are clearly mentioned below:

Diverse renewable, distributed energy sources, batteries, and electrical vehicles have been integrated into the multi-microgrids with the proposed CFOID-FOPIDN controller.

The effectiveness of the suggested CFOID-FOPIDN is ascertained to compare with other controllers such as PID/ID/FOID/FOPIDN/CFOID-FOPI.

The BMO approach is implemented to design the suggested controller as well as the productiveness of the BMO is examined with GA, PSO, CS, and Bat.

The amalgamation of DPFC and RFBs are employed coordinately with BMO optimized CFOID-FOPIDN controller for enriching the system performance.

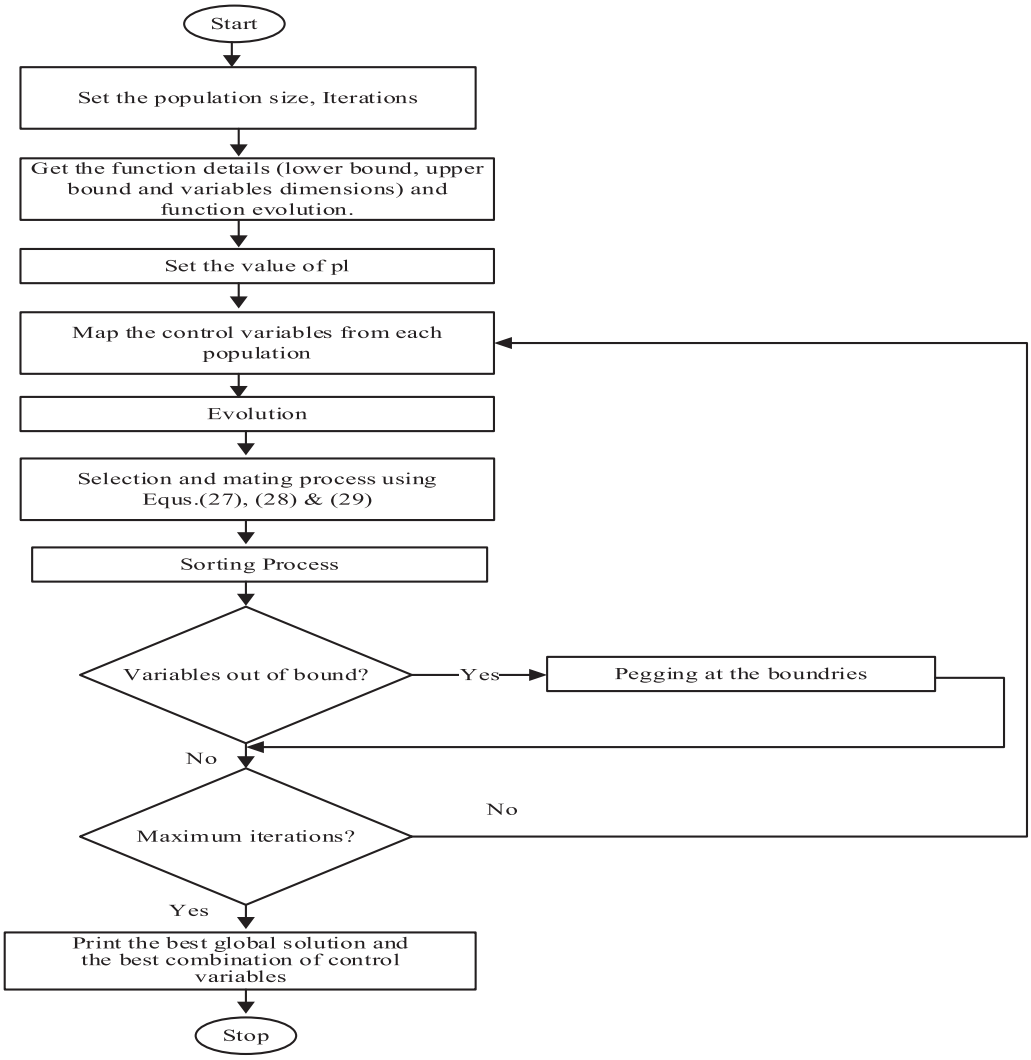


Fig. 2. Flowchart of BMO algorithm.

Eventually, the effectiveness of the proposed CFOID-FOPIDN controller is also investigated by distinct operating conditions, such as SLP with no disturbances in other sources, sudden change in wind speed, change in solar irradiance, and sudden changes in both wind and irradiance levels.

The remaining of this paper is coordinated as follows: Section 2 describes the modelling of multi-microgrid system with dissimilar sources. The structure of the CFOID-FOPIDN is discussed in Section 3. The discussion of BMO approach is shown in Section 4. Simulation outcomes and discussion of this work is demonstrated in Section 5. Finally, the concluding remarks of this work is exhibited in Section 6.

Modelling of multi-microgrid (MMG) system

In this study, two microgrids are considered and interconnected through tie-lines for the analysis with diverse renewable/distributed energy sources, and energy storage devices. The first microgrid (MG-I) consists of a diesel engine, FC, solar PV, and EV. Likewise, battery energy storage system (BESS), wind generator, diesel engine, and EV system has been taken in the second microgrid (MG-II) system. The parameters of the suggested system are represented in Appendix. Moreover, the proposed MMG system has been illustrated in Fig. 1. The suggested each MG has a centralized controller (CFOID-FOPIDN) and can be contributed the control signals to the diesel engine. Besides, the gains of the CFOID-FOPIDN in this work are optimized by the BMO approach for enriching the interconnected MMGs dynamic performance. Flowchart of whole control process is shown in Fig. S1. The detailed representation of each component in MMGs is explained as follows:

PV modeling

The electrical power supplied by the PV system is represented as [32,33]:

$$P_{pv} = \eta \varnothing_{Ir} S[1 - 0.005(T_a + 25)] \quad (1)$$

Where, η indicates the PV conversion efficiency, $\varnothing_{Ir}$ indicates the solar irradiance and measured in W/m^2. Also, the area of the PV array is represented with S and measure in m^2 and T_a is the ambient temperature (25 °C). Moreover, different types of power electronics devices like inverters are utilized for interfacing and synchronizing with AC microgrids.

The transfer function model of the PV is demonstrated by first-order lag as:

$$G_{PV}(s) = \frac{\Delta P_{PV}}{\varnothing_{Irr}} = \frac{K_{PV}}{1 + S.T_{PV}} \quad (2)$$

Where $K_{PV} = 1$ is the gain constant and $T_{PV} = 1.8$ is the time constant. Here, all uncertainties are not considered.

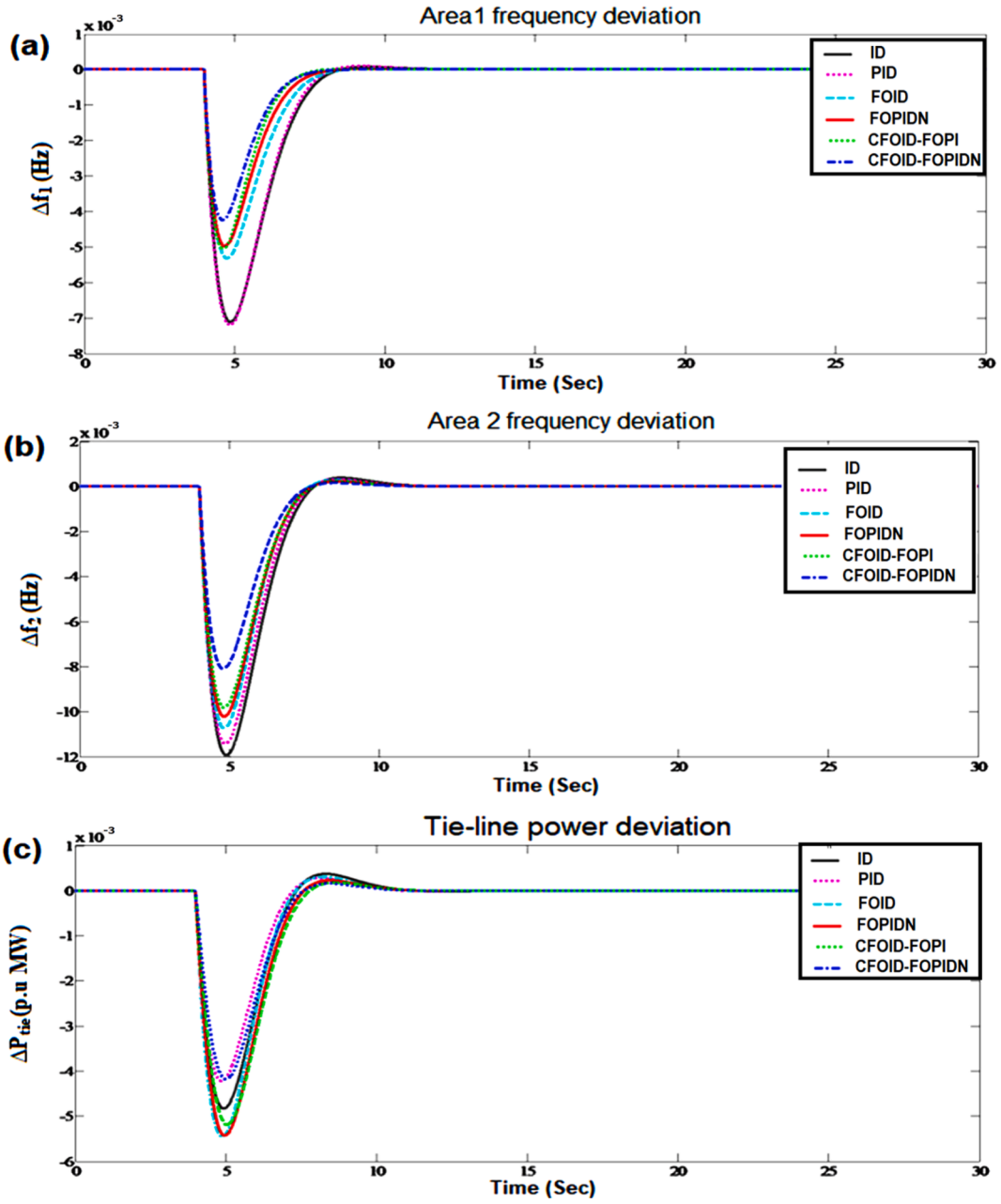


Fig. 3. Frequency and tie-line variation responses with the proposed and other controllers: (a) Δf_1, (b) Δf_2 and (c) ΔP_{tie}..

Wind turbine generator (WTG) modeling

The mechanical power of wind turbine is extremely deviating due to continuous variations in wind speed and represented as [32,33]:

$P_{Wind} = \frac{1}{2}\rho A_r C_P V_W^3$ (3)Here, air density is represented with ρ, which is identical to 1.25 kg/$m^{3,}$ and swept area of blades is indicated with A_r, which is identical to 1735 m^2. Moreover, wind velocity is shown with V_W and measured in m/s. C_P is the power coefficient and calculated as:

$$C_P = (0.44 - 0.0167\beta).\sin\left[\frac{\lambda - 3}{15 - 0.3\beta}\right] - 0.0184(\lambda - 3)\beta \quad (4)$$

Where, β indicates as blade pitch angle and identical to 0.1745. λ denotes as tip speed ratio ($R_{blade}\omega_{blade}/V_W$). R_{blade}andω_{blade} are demonstrates the turbine blades length and rotational of the blades respectively, which are equal to 23.5 m and 3.1428 m/s.

The model of WTG is illustrated by first-order lag as:

$$G_{WTG}(s) = \frac{P_{WTG}}{P_{Wind}} = \frac{K_{WTG}}{1 + S.T_{WTG}} \quad (5)$$

Diesel engine generator (DEG) modeling

In interconnected MGs, the DEG can be considered as a significant source for delivering uninterrupted and high-reliability power for critical loads as a result of continuous changes in outcomes of wind and PV systems. The DEG model is denoted by first-order lag as [32,33]:

$$G_{DEG}(s) = \frac{K_{DEG}}{1 + S.T_{DEG}} \quad (6)$$

Where, the gain and time constant are indicated as K_{DEG} and T_{DEG}, respectively. They are equal to 1 and 2, respectively.

Table 1
Performance of the proposed and other controllers.

Controllers	Profile	OS	US	ST
ID	ΔF_1	0	−0.0068	12.8
	ΔF_2	0	−0.118	13.2
	ΔP_{tie}	0.008	−0.0048	12.3
PID	ΔF_1	0	−0.0071	11.5
	ΔF_2	0	−0.107	12.7
	ΔP_{tie}	0.007	−0.0039	11.4
FOID	ΔF_1	0	−0.0052	9.8
	ΔF_2	0	−0.0098	11.5
	ΔP_{tie}	0.005	−0.0057	10.2
FOPIDN	ΔF_1	0	−0.0048	8.6
	ΔF_2	0	−0.0094	10.6
	ΔP_{tie}	0.004	−0.0056	9.9
CFOID-FOPI	ΔF_1	0	−0.0047	7.7
	ΔF_2	0	−0.0089	9.7
	ΔP_{tie}	0.002	−0.0053	9.5
CFOID-FOPIDN	ΔF_1	0	−0.0038	6.9
	ΔF_2	0	−0.0084	8.5
	ΔP_{tie}	0	−0.0065	8.9

Battery energy storage system (BESS) modeling

The BESS is performing a great role in multi-microgrid system for sustaining dynamic stability by delivering the power in a very less time. As the intermittent nature of wind and PV plants are integrated into the MGs, the BESS can be employed effectively while charging or discharging conditions. Several researchers are designed the models for BESS, which are more suitable to obtaining favourable outcomes with the incorporation of small disturbances. The BESS model is indicated by first-order lag as [32,33]:

$$G_{BESS}(s) = \frac{K_{BESS}}{1 + S.T_{BESS}} \tag{7}$$

Here, K_{BESS} and T_{BESS} are denoted as the gain and time constants of the BESS, respectively. Which are $K_{BESS} = -1/300 \text{and} T_{BESS} = 0.1$.

Electric vehicles (EVs) modeling

The electrical representation of EV can be shown in Fig. 1. From Fig. 1, the distributed grid has been interfaced with a bi-directional converter and other power electronic components. The EV battery delivered apparent power to the grid is calculated as [32,33]:

$$S_{EV} = V.I^{*} \tag{8}$$

Where, V represents the voltage at the receiving end, and I is delivered current from the battery. Similarly, the EV battery supplied active power to the grid is measured as (Equ.S1).

The controlling of a converter can be made by pulse width modulation (PWM); thus the reactive power of the EV keeps zero. In practical applications, the converter has been coupled with a grid through an inductor. In AGC studies, the EV can be illustrated with first-order transfer function. The time constant of EV is shown as (Equ. S2).

The transfer function modeling of EV is (Equ. S3).

Furthermore, the state of charge (SOC) of EV can be measured as (Equ. S4).

Fuel cell (FC) modeling

The FC is a substantial device for increasing the system dynamic stability at uncertainties in interconnected power systems. Since the integration of wind and PV systems in interconnected microgrids systems, the FC can be effectively compensated for the significant changes in power due to rapid changes in wind and PV outputs. The modeling of the FC can be shown by first-order lag as (Equ. S5) [32].

Modeling of distributed power flow controller (DPFC) and redox flow batteries (RFBs)

Many countries in this world are attempting to produce a vast amount of power with the incorporation of sustainable, renewable, distributed, and green energy sources. Nevertheless, the toughest duty in interconnected microgrids is to be sustained the proper operation and control as a result of continuous variations in atmospheric circumstances. Furthermore, the tie-lines of interconnected microgrid are generally congested in multi-areas with high penetration of distributed, renewable and green energy sources, which may degrade the ability of tie-lines. Hence, the interconnected microgrid system has to be upgraded with effective control techniques for enhancing the utilization of renewable power. The coordinative combination of energy storage devices (ESDs) and flexible alternating current transmission system (FACTS) approaches are essential to enriching the power flow capability, reduction in operational cost, and requisite provision of ancillary services [17]. Besides, the block diagrams of FACTS controller and ESD i.e. RFB are illustrated in Fig. S2(a) and (b), respectively. A sophisticated FACT devices called DPFC is adopted in this work with the coordination of RFBs for escalating the dynamic performance of MMGs. The modelling of the DPFC and RFB is illustrated via (Equ. S6) and (Equ.S7), respectively [17].

Design of CFOID-FOPIDN

Cascade combination of FOID and FOPIDN is utilized in this study for enriching the dynamic stability of the multi-area interconnected microgrid system under dissimilar operating conditions. Cascade controllers are more appropriate over traditional controllers to tracking the set-point and enhances the disturbance rejection. The structure of the cascade controllers has consisted of two control loops such as inner/secondary/slave and outer/primary/master loops. The output of the slave controller is the set point to the outer controller. Moreover, the slave controller can be contributed a fast response than the outer controller for rejecting the disturbance and avoid to the extent in other parts of the system. The benefit of the outer controller is to enhance the quality of the final outcome of the system. Other privileges of cascade controllers are to rejecting the load disturbance, strengthening the outcome, accurately manipulating the energy or mass, and compensating for the uncertainties [20–22]. The block diagram of cascade control scheme and structure of the suggested CFOID-FOPIDN are demonstrated in Fig. S3(a) and (b), respectively. In Fig. S3(a) there are R (S) and D(S) are representing the reference and disturbance inputs to the plant. Similarly, G_{C1} and G_{C2} are the transfer functions (TFs) of the primary and secondary controllers respectively as well as G_{P1}, G_{P2} illustrates the inner and outer TFs. Finally, Y(S) shows the outcome of the system. The TFs models of the suggested FOID and FOPIDN are illustrated as (Equ.S8 and Equ.S9).

In order to check the effectiveness of the FO controllers in closed loop system, it is compared with the integer order (IO) controllers using root locus plot as represented in Fig. S4. As per the plot in Fig. S4, the poles of the FO-based closed loop system move in the left half of s-plane as a result of stability of the system. For IO-based controllers, the system remains unstable for gain parameters.

Further, the cascade combination of FOID-FOPIDN has ten tuning parameters, which are $K_P/K_I/K_{I_1}/K_D/K_{D_1}$ and FO based operators are λ, λ_1, μ, μ_1, N. The FOID and FOPIDN give more extent of freedom, effective outcomes in terms of fewer oscillations/settling time/undershoots/overshoots, and high robustness in the presence of uncertainties/variations in parameters. These advantages are motivated to amalgamate the FOID and FOPIDN in a cascaded structure.

The FOID and FOPIDN are the further extension of traditional ID and PIDN controllers and constructed through fractional calculus. In order to perform the FO transfer function (TF) in real-time or simulation works, the typical method IO TF is used to approximate the FO TF. The Crone

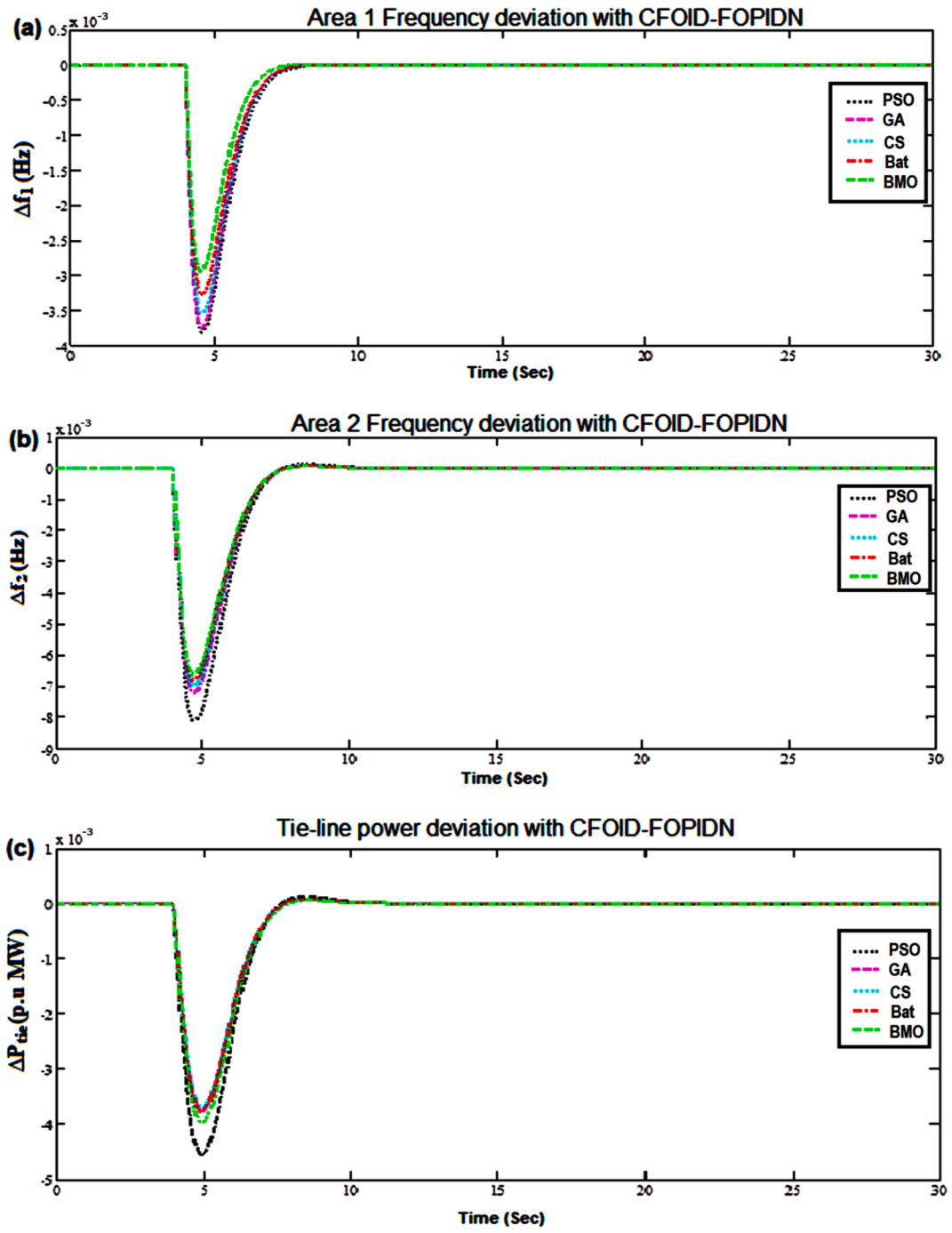


Fig. 4. Frequency and tie-line responses with diverse optimization methods: (a) Δf_1, (b) Δf_2 and (c) ΔP_{tie}.

Table 2
The statistical analysis of the controller with various optimization methods.

Controller	Response	Parameter	PSO	GA	CS	BA	BMO
CFOID-FOPIDN	ΔF_1	OS	0	0	0	0	0
		US	–0.0038	–0.0035	–0.0034	–0.0032	–0.0028
		ST	6.7	6.2	5.8	5.6	5.2
	ΔF_2	OS	0	0	0	0	0
		US	–0.0082	–0.0072	–0.0069	–0.0067	–0.0063
		ST	8.3	7.9	6.8	5.9	5.3
	ΔP_{tie}	OS	0	0	0	0	0
		US	–0.0048	–0.0039	–0.0035	–0.0036	–0.0037
		ST	9.2	8.8	7.5	6.8	5.6
		ITAE	0.715	0.638	0.614	0.590	0.542

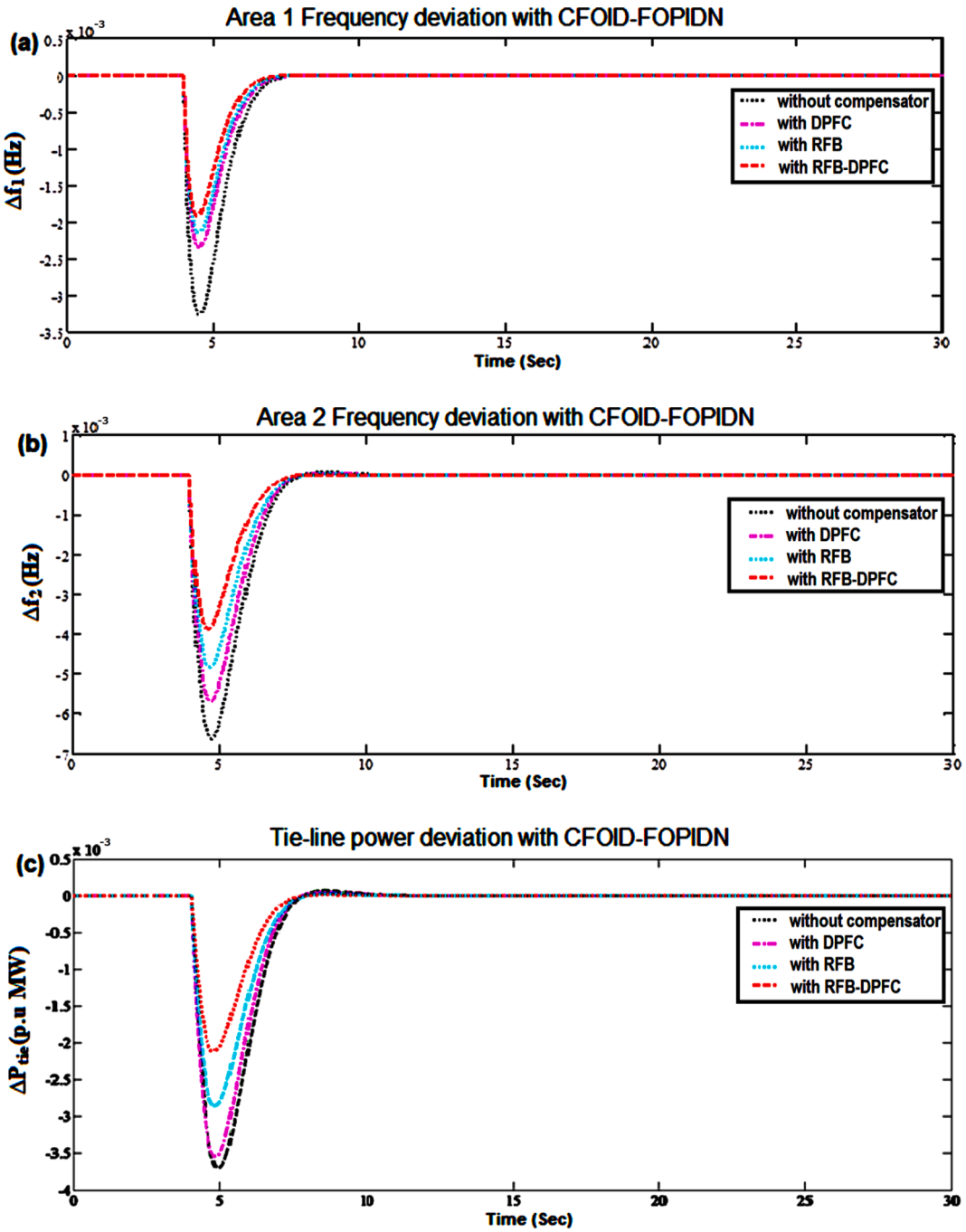


Fig. 5. Changes in frequency and tie-line power with compensators: (a) Δf_1, (b) Δf_2 and (c) ΔP_{tie}.

Table 3
Performance analysis of the controller with only compensator.

Controller	Response	Parameter	Without compensator	With DPFC	With RFB	DPFC-RFB
CFOID-FOPIDN	ΔF_1	OS	0	0	0	0
		US	−0.0034	−0.0024	−0.0022	−0.0018
		ST	6.9	6.4	5.8	5.6
	ΔF_2	OS	0	0	0	0
		US	−0.0068	−0.0059	−0.0051	−0.0037
		ST	8.5	6.7	6.5	5.5
	ΔP_{tie}	OS	0	0	0	0
		US	−0.0035	−0.0032	−0.0029	−0.0021
		ST	8.9	7.7	7.4	6.2

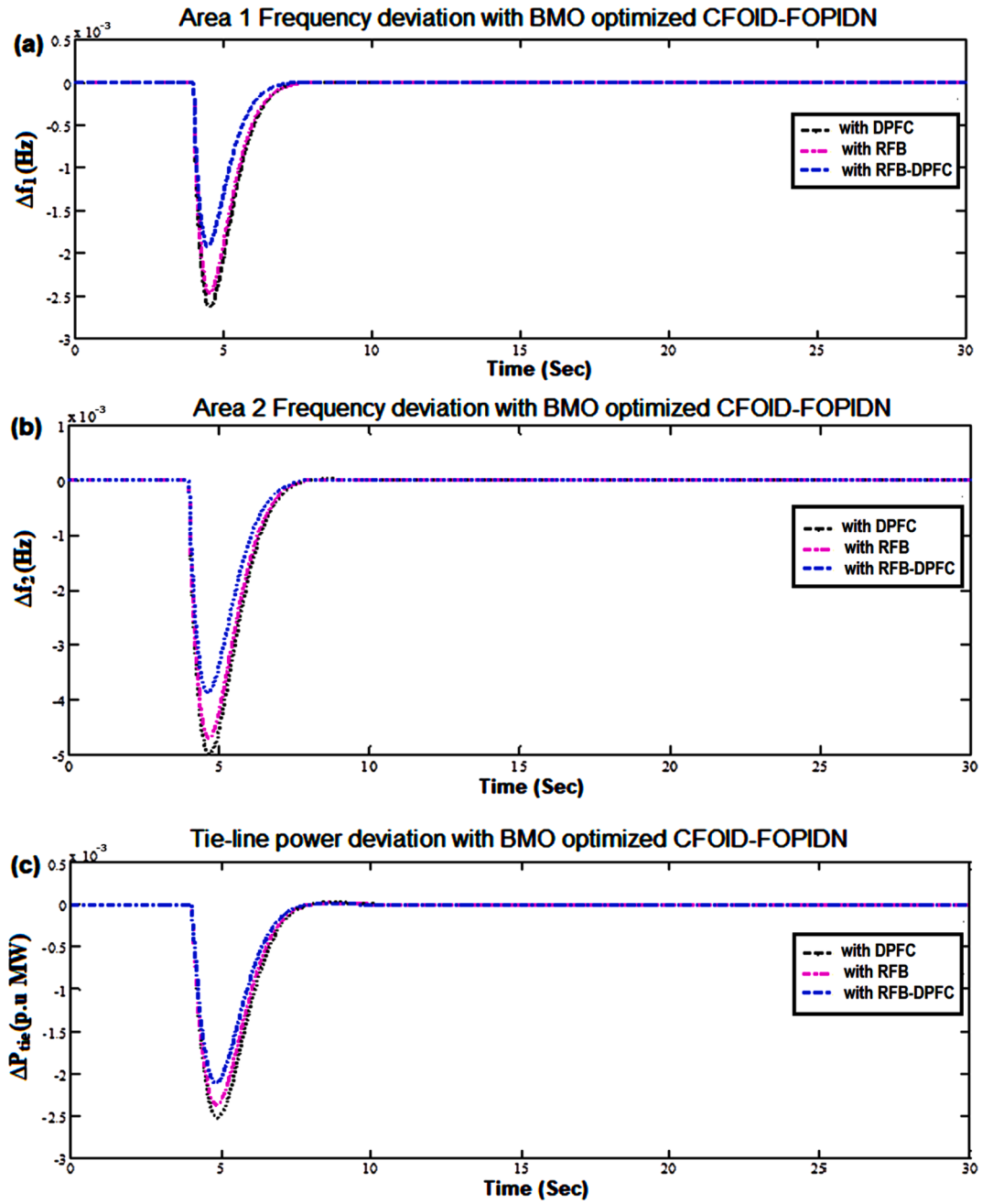


Fig. 6. Frequency and tie-line power deviations with BMO-based compensators: (a) Δf_1, (b) Δf_2 and (c) ΔP_{tie}.

Table 4
Dynamic analysis of the BMO-based CFOID-FOPIDN with compensator.

Method	Profile	OS	US	ST	ITAE
BMO-CFOID-FOPIDN-DPFC	ΔF_1	0	−0.0027	5.3	0.470
	ΔF_2	0	−0.0051	5.27	
	ΔP_{tie}	0	−0.0025	5.4	
BMO-CFOID-FOPIDN-RFB	ΔF_1	0	−0.0024	5.2	0.422
	ΔF_2	0	−0.0046	5.28	
	ΔP_{tie}	0	−0.0023	5.45	
BMO-CFOID-FOPIDN-RFB-DPFC	ΔF_1	0	−0.0017	5.1	0.314
	ΔF_2	0	−0.0035	5.12	
	ΔP_{tie}	0	−0.0021	5.3	

approximation has been utilized in this study and provided by Oustaloup. It offers an appropriate TF to approximate the FO derivatives and integrators using a recursive distribution of N number of zeros/poles. Which are represented by (Equ. S10, Equ.S11, Equ.S12). In this work, for FOID and FOPIDN controller, fifth order Oustaloup's rational recursive approximation is reported in the frequency range as $[10^{-2}, 10^{2}]$ rad/s, and Na value is 6. The output of the proposed CFOID-FOPIDN is the control input (ΔP_C) and expressed as (Equ.S13).

In this study, the effectiveness of the CFOID-FOPIDN controller is also examined by diverse operating conditions, such as step load perturbation (SLP) with no disturbances, sudden change in wind speed, change in solar irradiance with no variations in other sources.

Table 5
Gain values of the proposed CFOID-FOPIDN controller.

Compensator	Profile	CFOID-FOPIDN gains	BMO
DPFC-RFB	Area 1	K_I	2.736
		K_D	1.725
		λ	0.721
		μ	0.892
		K_P	1.625
		K_{I_1}	2.758
		K_{D_1}	1.684
		N	94.27
		λ_1	0.684
		μ_1	0.852
	Area 2	K_I	1.851
		K_D	0.344
		λ	0.521
		μ	0.692
		K_P	1.383
		K_{I_1}	1.684
		K_{D_1}	0.297
		N	75.28
		λ_1	0.484
		μ_1	0.752

Problem formulation

The cascade amalgamation of FOID and FOPIDN is employed in this MMG system as a centralised controller for dwindling the changes in tie-line and system frequency. Nevertheless, an effective and substantial optimization method is needed to be obtained the ideal gains of CFOID-FOPIDN for regulating the changes in interconnected MMG systems. Hence, a new bio-inspired barnacle mating optimizer (BMO) algorithm is developed to tune the gains ($K_P/K_I/K_{I_1}/K_D/K_{D_1}/\lambda/\lambda_1/\mu/\mu_1/N$) of the CFOID-FOPIDN. Distinct types of objective functions (J) are commonly utilised in the industry for modelling the controllers. The objective functions are integral of squared error (ISE), integral of absolute error (IAE), time multiplied ISE (ITSE), and time multiplied IAE (ITAE). The ITAE is used as the objective function in this study because of their substantial and preeminent performance and shown in (Equ.S14) [19]. The CFOID-FOPIDN approach optimization duty is indicated via constrained optimization problem. Minimize J subject to Equ. S15. From Equ.S15, the minimum and maximum limits of the suggested CFOID-FOPIDN is represented with min. and max. respectively. The min. and max. limits of $K_{P_i}^{min}$ is [0 2], [0 4] for K_i^{min} and K_{1i}^{min}, [0 2] for $K_{D_i}^{min}$ and K_{D1i}^{min}, [0 1] for λ_i^{min} and λ_{1i}^{min}, [0 2] for μ_i^{min} and μ_{1i}^{min} and [0 500] for N_i^{min}. Further, the assumed maximum frequency and tie-line power deviations are 1 to 1.2 % (scheduled value) and 5 % (total capacity) respectively during the optimization for perturbations.

BMO approach

Barnacle

Barnacles are typically micro-organisms, which are endured since Jurassic times. They have determined in seawater or very proximately to seawater. Barnacles can swim in water from the birth stage, and start themselves to attach the shell, rocks, or other objects in the water when they come into the adult period. The majority of barnacles are hermaphroditic, which means they reproduce both male and female. Around 1400 kinds of barnacles can be found in seawater, with acorn barnacles being the most frequent [31–34].

One of the interesting aspects of the barnacles is that they have the longest penis, it can be grown up to seven to eight times the body length by hermaphroditic filter-feeders. All the neighbour mates must be in the range of their penis for a group of barnacles mating. Moreover, they have a great capacity to alter the shape and size of their penises to proximately match local wave circumstances. In the light of lower wave action, a longer penis can be reached more mates. For higher wave action, a shorter penis is more movable inflow and hence can meet more mates. The life cycle of the barnacles has consisted of three stages such as (a) a pelagic, suspension-feeding nauplius larva, (b) a pelagic, non-feeding cypris larva, and (c) a benthic suspension feeding-adult.

Other significant attributes of the isolated barnacles are sperm casting and self-fertilization. Self-fertilization may be theoretically viable, but it is difficult to achieve in practice. Sperm-cast mating is another process to fertilize the eggs, however, this mating process can be done by releasing the sperm into the water. Barnacle mating behaviour is being used as inspiration in the enhancement of BMO for determining optimization challenges.

Hardy-Weinberg principle

According to the Hardy-Weinberg principle, genetic changes in a population will remain consistent from generation to generation. When the process of mating has random nature in a high population with no disruptive situations, then the principle states that the frequencies of allele and genotype should be constant due to they are in equilibrium.

Two alleles such as D (Dad) and M (Mom) are considered as well as their frequencies can be shown as f(D) = p and f(M) = q, respectively. For normal mating, the genotype frequencies can be demonstrated as f (DD) = p^2 for DD homozygotes, f(MM) = q^2 for MM homozygotes and f (DM) = 2pq for DM heterozygotes. The total of all entries is $p^2 + q^2 + 2pq = 1$. It's also obvious that $p + q = 1$. As a result, the generation of innovative offspring is dependent on the barnacle's parents' p and q. and these are discussed in the following sub-sections.

C. BMO development

- *Initialization.*

First, initialize the individuals (barnacles) for the solution of the candidate and shown in the matrix as Equ. S16, Equ. S17, Equ. S18. In this stage, after ascertaining the evaluation of the objective function for every barnacle, it has been sorted and saved at the top of vector X as a first iteration solution.

- *Selection process.*

Barnacles' mate selection procedure is mostly depending on the length of their penis, pl. The assumptions that guide this selection procedure are as follows.

1. Although the selection process appears to be random, it will be limited to the barnacle's penis length, pl.
2. Each barnacle has the ability to gather or deliver sperm to other barnacles. In the BMO approach, however, each barnacle may only be fertilized by one barnacle at a time.
3. If at a particular point, the same barnacle (self-mating) is chosen by the selection process. Though barnacles have both male and female reproductions, self-matting or self-fertilization is very rare due to the maximum number of barnacles are not self-mating [31,34]. As a result, self-mating will not be considered in this work, and the process will continue without a new generation of offspring.
4. The sperm cast process will take place if the selection at a certain iteration is greater than the determined pl.

Based on the above assumptions, the BMO includes the process of exploration and exploitation. The offspring generation is progressed by the sperm cast mechanism (exploration). It can be expressed in mathematical functions and represented by Equ. S19 and Equ. 20. The Equ. S19 and Equ. S20 demonstrated that the selection process has been presented in random nature and achieved the first considered assumption.

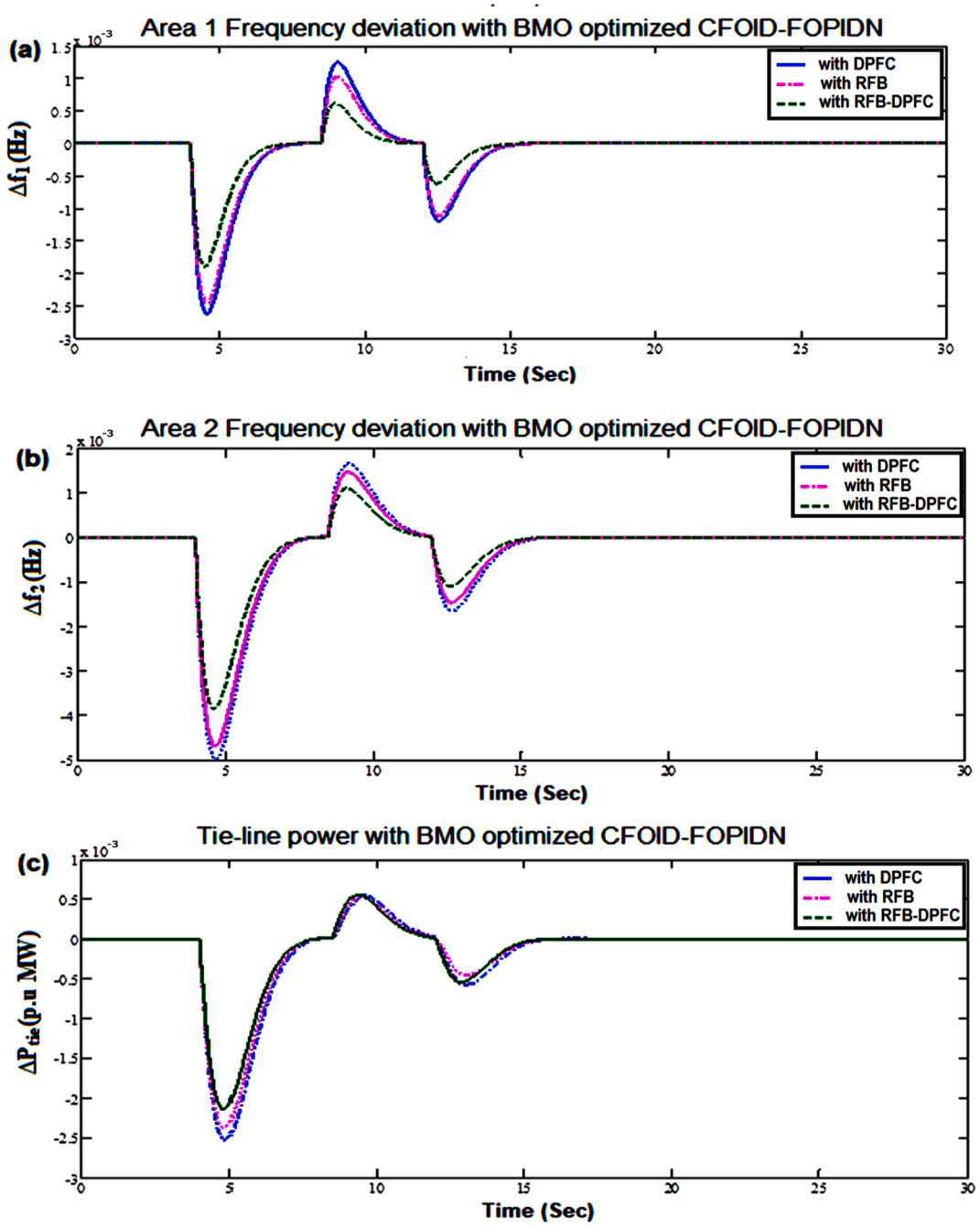


Fig. 7. Frequency and tie-line variations under fast changes in wind speed: (a) Δf_1, (b) Δf_2 and (c) ΔP_{tie}.

Table 6
Numerical values of ST, OS, US & ITAE under wind variations.

SLP & Wind	Controller	Compensator	Profile	OS	US	ST	ITAE
SLP = 0.2 at t = 4 s, ΔV_ω = 7.5 to 12 m/s at t = 10 s, 12 to 10 m/s at t = 15 s	CFOID-FOPIDN	DPFC	ΔF_1	0.0013	−0.0027	7.9	0.464
			ΔF_2	0.0018	−0.005	8.4	
			ΔP_{tie}	0.00059	−0.0025	9.2	
		RFB	ΔF_1	0.0011	−0.0025	7.6	0.440
			ΔF_2	0.0016	−0.0048	8.1	
			ΔP_{tie}	0.00058	−0.0024	8.8	
		RFB-DPFC	ΔF_1	0.0006	−0.0018	6.2	0.344
			ΔF_2	0.0009	−0.0039	7.3	
			ΔP_{tie}	0.00060	−0.0021	7.8	

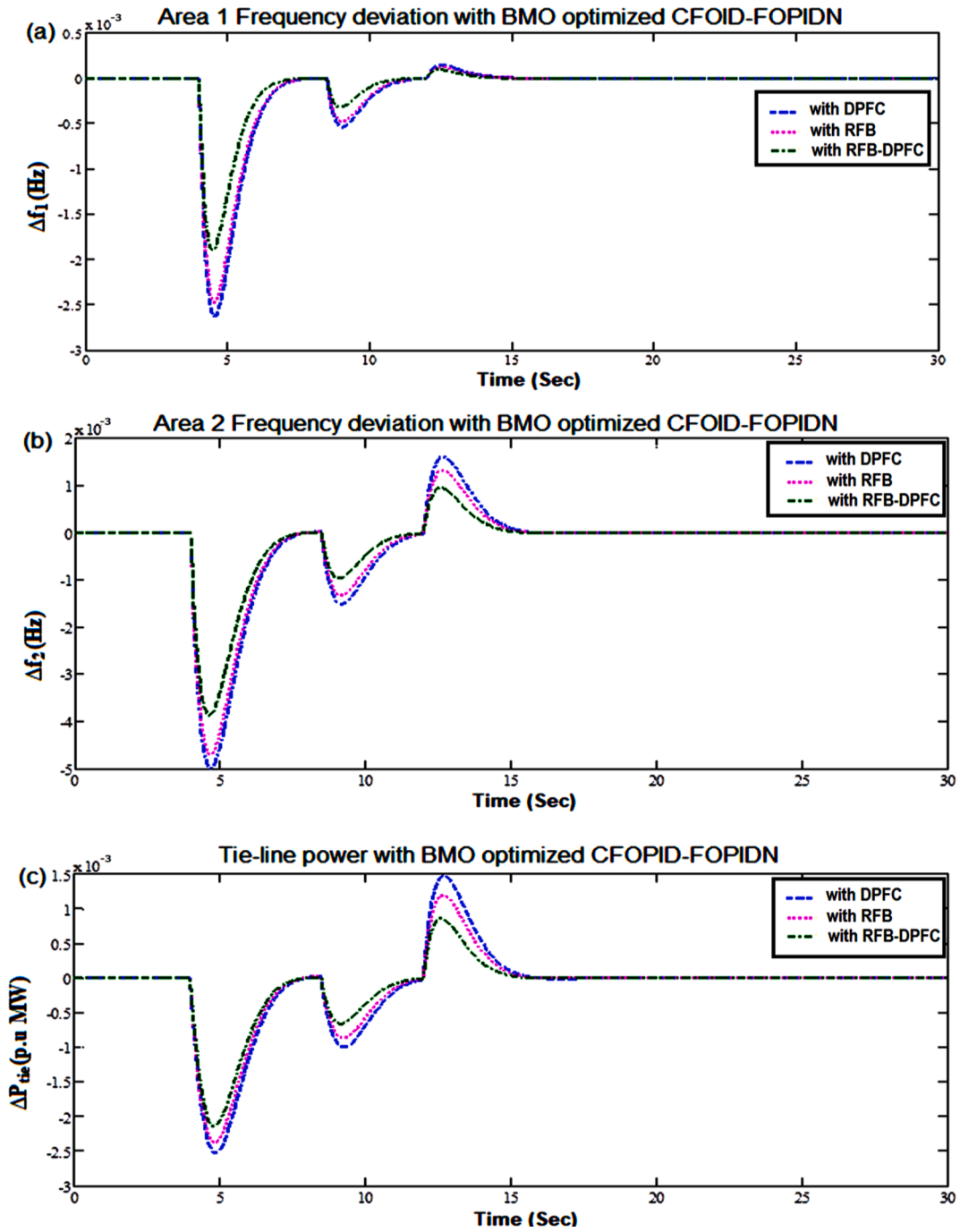


Fig. 8. Frequency and tie-line power deviations under sudden changes in solar irradiance: (a) Δf_1, (b) Δf_2 and (c) ΔP_{tie}.

Table 7
Performance analysis of the compensators with CFOID-FOPIDN under sudden changes in solar irradiance.

SLP & Solar irradiance	Controller	Compensator	Profile	OS	US	ST	ITAE
SLP = 0.2 at t = 4 s, Δϕ = 500 to 300 W/m^2 at t = 10 s, 300 to 700 W/m^2 at t = 15 s	CFOID-FOPIDN	DPFC	ΔF_1	0	−0.0025	8.5	0.452
			ΔF_2	0	−0.005	8.9	
			ΔP_{tie}	0	−0.0026	8.7	
		RFB	ΔF_1	0	−0.0023	8.2	0.428
			ΔF_2	0	−0.0048	8.7	
			ΔP_{tie}	0	−0.0024	8.5	
		RFB-DPFC	ΔF_1	0	−0.0017	6.8	0.320
			ΔF_2	0	−0.0036	7.9	
			ΔP_{tie}	0	−0.0021	8.0	

- *Reproduction.*

The process of reproduction is suggested in BMO, which is slightly dissimilar as compared with other evolutionary optimization methods. As there are no particular equations/mathematical formulas for ascertaining the barnacle's reproduction, the BMO approach is primarily depending on the genotype frequencies or their inheritance characteristics for generating the offspring based on the principle of Hardy-Weinberg. For producing the new offspring from barnacle's parents, the following expression is suggested in Equ. S21.

It is worth mentioning that the pl value may show a significant role in ascertaining the exploitation and exploration process. Since the selective barnacles are to be mated when the penis length (pl) of dad's barnacle is in a suitable range, the exploitation process is happened (Equ. S21). In this algorithm, the sperm cast is served as an exploration process when the selection of barnacles to be mated exceeds the pl value. The sperm cast process can be exhibited as Equ. S22. The mom barnacle can be generated the new offspring for the exploration process. The flowchart of the BMO is demonstrated in Fig. 2. To control the solution of expansion of the matrix as a function of population size, the new offsprings are assessed and combined with the parents. Then, the sorting process is made for selecting the 50 % top solution that fits the population size. The poor outcomes are considered to be removed and dead. The pseudo code of the BMO is mentioned below:

Initialize the population size, X_i

Evaluate the fitness of every barnacle
Sorting to locate the best value at the top of population
(T = Best solution)
While (I < Maximum iterations)
Set the pl value
Selection using Equ.(27 & 28)
If selection of Dad and Mom = pl
for each variable
Offspring generation using Equ(29)
end for
else if selection of Dad and Mom > pl
for each variable
Offspring generation using Equ(30)
end for
end if
Bring the current barnacle back if it goes outside the range
Calculate the fitness of each barnacles
Sorting and upgrade T if there is better outcome.
I = I + 1
end while
Return T.

Results and discussion

For assessment of the proposed study, a MMG system has been examined with various time-domain specifications like settling time (ST), overshoot (OS), and undershoot (US) under distinct perturbation and operational circumstances. The proposed MMG system is developed in MATLAB/Simulink software. To show the effectiveness of the suggested CFOID-FOPIDN, its outcomes are analysed with PID/ID/FOID/FOPIDN/CFOID-FOPI controllers. Subsequently, the superiority of the BMO-based CFOID-FOPIDN controller is verified over GA, PSO, CS, and Bat algorithms-based CFOID-FOPIDN controller. Moreover, the amalgamation of RFBs and DPFC is utilized with proposed technique for ameliorating the frequency and tie-line responses. Further, to evaluate the effectiveness of the proposed CFOID-FOPIDN, the outcomes of the MMGs are compared with the PID/ID/FOID/FOPIDN/CFOID-FOPI controllers under distinct operating/renewable/loading conditions as represented below via different case studies:

Case 1 SLP with no disturbances in considered renewable sources.
Case 2 variations of wind speed.
Case 3 Variations of irradiance.
Case 4 variations of both wind speed and irradiance.

A. Case 1.

In this case, solar irradiance (ϕ), average wind speed (V_ω) and temperature (T_a) are considered as 500 W/m^2, 7.5 m/s, and 25 °C, respectively for evaluating the system performance with the proposed control approach. Further, SLP is taken as 0.2 pu and it has been used in microgrid-1 at t = 4.0 s for examining the suggested MMG system. Moreover, other parameter settings of these simulation outcomes are population of barnacles = 10, pl = 7 and maximum iteration = 40. Fig. 3 (a-c) demonstrates the frequency variations in area-1, area-2 and changes in tie-line power with ID/PID/FOID/FOPIDN/CFOID-FOPI/CFOID-FOPIDN controllers.

It is observed from the inspection of Table 1 that for area-1, the settling times with the utilized controllers are 6.9 s for CFOID-FOPIDN, 7.7 s for CFOID-FOPI, 8.6 s for FOPIDN, 9.8 s for FOID, 11.5 s for PID and 12.8 s for ID. For area-2, 8.5 s for CFOID-FOPIDN, 9.7 s for CFOID-FOPI, 10.6 s for FOPIDN, 11.5 s for FOID, 12.7 s for PID, and 13.2 s for ID. Similarly, for tie-line power, 8.9 s for CFOID-FOPIDN, 9.5 s for CFOID-FOPI, 9.9 s for FOPIDN, 10.2 s for FOID, 11.4 s for PID and 12.3 s for ID. As per the numerical simulation outcomes, the cascade combination of FOID and FOPIDN (CFOID-FOPIDN) provides the finest settling time and improved the system dynamic response as compared with other studied controllers. The numerical values of the developed and other controllers are mentioned in Table 1.

Subsequently, five optimization approaches, such as PSO/GA/CS/BA/BMO are employed in this work for attaining the suitable gains of the CFOID-FOPIDN controller. The frequency deviations of area-1, area-2 and tie-line power variance are illustrated in Fig. 4(a-c). From Fig. 4(a), BMO-based CFOID-FOPIDN exhibits less settling time (5.2 s) as contrasted with other methods like 5.6 s for BA, 5.8 s for CS, 6.2 s for GA and 6.7 s for PSO. Fig. 4(b) is the area-2 frequency response, and its settling times are 5.3 s for BMO, 5.9 s for BA, 6.8 s for CS, 7.9 s for GA and 8.3 s for PSO. In Fig. 4(c), the settling times for tie-line power are 5.6 s for BMO, 6.8 s for BA, 7.5 s for CS, 8.8 s for GA and 9.2 s for PSO. The simulation results disclosed that the BMO optimized CFOID-FOPIDN stabilizes the deviations in frequency and tie-line more quickly and ameliorated the system dynamic stability hastily in comparison to other methods. The statistical analysis of the controller with various optimization approaches is noted in Table 2. The time and performance analysis of the controller with various optimization methods are presented in Fig. S5(a-b). From Table 2 and Fig. S5(a-b), it is observed that BMO-based CFOID-FOPIDN controller in superior to other alternatives in terms of less values of ITAE and ST (except for ΔP_{tie}).

To amplify the system dynamic performance at perturbations, the compensators such as DPFC and RFBs are utilized separately and jointly with CFOID-FOPIDN controllers. Fig. 5(a-c) shows the frequency variation in area-1, area-2 and tie-line power with RFB, DPFC, and RFB-DPFC in the presence of the CFOID-FOPIDN controller. The settling times for area-1 are 5.6 s for DPFC-RFB, 5.8 s for RFB, and 6.4 s for DPFC. Similarly, for area-2, 5.5 s for DPFC-RFB, 6.5 s for RFB and 6.7 s for DPFC. For tie-line power, 6.2 s for DPFC-RFB, 7.4 s for RFB and 7.7 s for DPFC. Hence, amalgamation of RFB and DPFC with CFOID-FOPIDN contributes the better enrichment for suppressing the variances in frequency and tie-line power. Performance analysis of the controller with/without compensator is indicated in Table 3.

Conversely, the BMO algorithm is presented to optimize the CFOID-FOPIDN with compensators for improving the dynamic stability and is represented in Fig. 6(a-c). The settling times of area-1 is 5.1 s for DPFC-RFB, 5.3 s for DPFC and 5.20 s for RFB. Whereas, in area-2, 5.12 s for DPFC-RFB, 5.27 s for DPFC and 5.28 s for RFB. For tie-line power, 5.3 s about to DPFC-RFB, 5.4 s about to DPFC and 5.45 s about to RFB. Moreover, the ITAE values of DPFC-RFB (ITAE = 0.314) is also less as compared with DPFC (ITAE = 0.470) and RFB (ITAE = 0.422). The reduction level of the objective function (ITAE) with DPFC-RFB is 25.5 % and 49.6 % as compared with RFB, DPFC, respectively. It should be noted that the joint operation of DPFC and RFB with BMO-based CFOID-FOPIDN demonstrates the supremacy than individual compensators for

ameliorating the dynamic performance. The dynamic analysis of the BMO-based CFOID-FOPIDN with a compensator is recorded in Table 4. The gain values of the proposed CFOID-FOPIDN is recorded in Table 5.

B. Case 2.

In this case, SLP is considered as 0.2 pu and applied to MG-1 at t = 4 s. Moreover, the average wind speed (V_ω) is swiftly increased from 7.5 to 12 m/s in a step at t = 10 s. Nonetheless, the increased wind speed is dropped suddenly from 12 to 10 m/s at t = 15 s, and does not consider any changes in solar irradiance ($\Delta\phi = 0$) and temperature ($\Delta T_a = 0$). Fig. 7(a-c) exhibits the frequency and tie-line variations under fast changes in wind speed ($\Delta V_\omega = \pm 4$ m/s) with suggested control methods.

The settling times of the area-1 are 6.2 s for DPFC-RFB, 7.6 s for RFB, and 7.9 s for DPFC. For area-2, 7.3 s for DPFC-RFB, 8.1 s for RFB and 8.4 s for DPFC, as well as tie-line powers are 7.8 s for DPFC-RFB, 8.8 s for RFB and 9.2 s for DPFC. The objective function (ITAE) of the DPFC-RFB is 0.344, 0.440 for RFB and 0.464 for DPFC. The mitigation rates (ITAE) of the DPFC-RFB are 27 % and 34 % as compared with RFB and DPFC respectively. The simulation outcomes of this investigation show that the proposed BMO-based CFOID-FOPIDN yields the finest performance with DPFC-RFB than other approaches under quick variations in wind speed ($\Delta V_\omega = \pm 4$ m/s). The ST, OS, and US values are reported in Table 6.

C. Case 3.

Under this case, SLP is considered as 0.2 pu, which is applied to MG-1 at t = 4. The sudden drop in solar irradiance (ϕ) from 500 to 300 W/m^2 at t = 10 s and quickly increase from 300 to 700 W/m^2 at t = 15 s with no changes in wind speed ($\Delta V_\omega = 0$) and temperature ($\Delta T_a = 0$) is considered. Deviations in frequencies of areas 1–2 and tie-line power deviation responses with proposed controllers under this condition are illustrated in Fig. 8(a-c). From the Fig. 8(a), the settling times in area-1 are 6.8 s, 8.2 s and 8.5 s for DPFC-RFB, RFB and DPFC, respectively. Fig. 8(b) illustrates the area-2 frequency deviations, and their values are 7.9 s for DPFC-RFB, 8.7 s for RFB and 8.9 s for DPFC. For tie-line power in Fig. 8(c), the values are 8.0 s for DPFC-RFB, 8.5 s for RFB and 8.7 s for DPFC. Additionally, the ITAE values for DPFC-RFB, RFB, and DPFC are 0.320, 0.428, 0.452, respectively. The decrement levels of ITAE with DPFC-RFB are 33.75 % and 41.25 % as compared with RFB, and DPFC. respectively. The numerical analysis of these controllers is represented in Table 7. As per simulation outcomes, the suggested CFOID-FOPIDN with DPFC-RFB contributes a low-rate deviation in frequency and tie-line as compared to studied methods under sudden changes in irradiance.

D. Case 4.

Regarding this case, the SLP (0.2 pu) is applied to the MG-1 at 4 s and varied both wind and irradiance levels for investigating the controller performance with FACTS and energy storage device. For wind speed, it is raised from 7.5 to 12 m/s at 10 s and dropped from 12 to 10 m/s at 15 s. Similarly, for solar irradiance, it is dropped from 500 to 300 W/m^2 at 10 s and raised from 300 to 700 W/m^2 at 15 s. Fig. S6(a-c) shows the frequency deviations in area-1, area-2 and tie-line variation, respectively. As per the simulation outcomes, the amalgamation of RFB and DPFC with BMO optimized CFOID-FOPIDN controller ensures the finest enhancement in reducing the changes of frequency and tie-line power. Further, it should be noted that the settling time, overshoot and undershoots of the Fig. S6(a-c) are decreased as compared with the case 3 (Fig. 8(a-c)) and case 2 (Fig. 7(a-c)) responses.

Conclusion

This work is undertaken to design a multi-microgrid system with the consideration of traditional/renewable/sustainable sources. Further, a new fractional order-based cascaded controller, such as CFOID-FOPIDN is proposed and implemented as a secondary controller for suppressing the variations in frequency and tie-line power. A newly developed barnacle mating optimizer is presented in this study to attain the suitable gains of the CFOID-FOPIDN controller. In this investigation, the validity of the proposed algorithm is evaluated and assessed with four optimization methods namely, particle swarm optimization, genetic algorithm, cuckoo search and bat algorithm. The results of this investigation show that the barnacle mating optimizer-based CFOID-FOPIDN yields better improvement in minimizing the settling time, overshoot and undershoot. Subsequently, the amalgamation of distributed power flow controller and redox flow batteries are incorporated into the microgrid system for mitigating the disparities in tie-line and enriches the dynamic stability. The evidence from this study suggests that the barnacle mating optimizer-based CFOID-FOPIDN with the combination of distributed power flow controller and redox flow batteries provides a low fluctuation rate in frequency and tie-line. Eventually, the effectiveness of the CFOID-FOPIDN controller is verified by diverse operating conditions with only redox flow battery, distributed power flow controller and coordination of both redox flow batteries and distributed power flow controller. Further work might investigate with the consideration of physical constraints such as governor dead band, generation rate constraint, and time-delay. Moreover, hybrid algorithms are required to be examined the cascade controllers for multi-microgrid systems.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This Project is supported by Universiti Malaysia Pahang (UMP) through grant no. (PGRS200322) and Peddakapu Kurukuri would like to acknowledge the Postdoctoral Research Fellowship awarded by the Centre for Advanced Industrial Technology, Universiti Malaysia Pahang.

Appendix

System Specifications	f = 60 Hz, Each area power rating = 1200 MW, Base power = 1200 MW
Multi-Microgrid details	$K_{PS} = 120$ Hz/MW. pu, $T_{PS} = 6$ s $R_1 = R_2 = 2.4$ Hz/puMW, $B_1 = B_2 = 0.425$ pu MW/Hz, $T_{PV} = 1.8$ s, $K_{PV} = 1$, $T_{WTG} = 1.5$ s, $K_{WTG} = 1$, $T_{DEG} = 1$ s, $K_{DEG} = 1$, $T_{BESS} = 0.1$ s, $K_{BESS} = -1/300$, $T_{EV} = 0.2$ s, $K_{EV} = -1/100$, $T_{FC} = 4$ s, $K_{FC} = -1/5$, $2\pi T_{12} = 0.545$ pu MW/Hz, $T_{DPFC} = 0.008$ s, $K_{DPFC} = 1$, $T_{RFB} = 0$ s, $K_{RFB} = 0.6768$.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.seta.2022.102823.

References

[1] Lekvan AA, Habibifar R, Moradi M, Khoshjahan M, Nojavan S, Jermsittiparsert K. Robust optimization of renewable-based multi-energy micro-grid integrated with flexible energy conversion and storage devices. Sustain Cities Soc 2021;64:102532.

[2] Sarangi S, Sahu BK, Rout PK. Distributed generation hybrid AC/DC microgrid protection: a critical review on issues, strategies, and future directions. Int J Energy Res 2020;44(5):3347–64.

[3] Karimi H, Jadid S, Makui A. Stochastic energy scheduling of multi-microgrid systems considering independence performance index and energy storage systems. J Energy Storage 2021;33:102083.

[4] Bazmohammadi N, Tahsiri A, Anvari-Moghaddam A, Guerrero JM. A hierarchical energy management strategy for interconnected microgrids considering uncertainty. Int J Electr Power Energy Syst 2019;109:597–608.

[5] Nayak PC, Prusty UC, Prusty RC, Panda S. Imperialist competitive algorithm optimized cascade controller for load frequency control of multi-microgrid system. Energy Sources, Part A Recover Util Environ Eff 2021:1–23.

[6] Peddakapu K, Mohamed MR, Sulaiman MH, Srinivasarao P, Veerendra AS, Leung PK. Performance analysis of distributed power flow controller with ultra-capacitor for regulating the frequency deviations in restructured power system. J Energy Storage 2020;31:101676.

[7] Yin L, Yu T, Yang B, Zhang X. Adaptive deep dynamic programming for integrated frequency control of multi-area multi-microgrid systems. Neurocomputing 2019; 344:49–60.

[8] Zeng GQ, Xie XQ, Chen MR, Weng J. Adaptive population extremal optimization-based PID neural network for multivariable nonlinear control systems. Swarm Evol Comput 2019;44:320–34.

[9] Chen B, Li Y, Li P, Wu D. Decentralized coordinated dispatch of multi-microgrid integrated energy system considering source-load uncertainty. IOP Conf Ser: Earth Environ Sci 2020;555(1):12105.

[10] Siti MW, Tungadio DH, Sun Y, Mbungu NT, Tiako R. Optimal frequency deviations control in microgrid interconnected systems. IET Renew Power Gener 2019;13(13): 2376–82.

[11] Fei J, Chen Y. Dynamic terminal sliding-mode control for single-phase active power filter using new feedback recurrent neural network. IEEE Trans Power Electron 2020;35(9):9906–24.

[12] Kasireddy I, Nasir AW, Singh AK. IMC based controller design for automatic generation control of multi area power system via simplified decoupling. Int J Control Autom Syst 2018;16(3):994–1010.

[13] Arya Y, Kumar N. Fuzzy gain scheduling controllers for automatic generation control of two-area interconnected electrical power systems. Electr Power Components Syst 2016;44(7):737–51.

[14] Chen MR, Zeng GQ, Xie XQ. Population extremal optimization-based extended distributed model predictive load frequency control of multi-area interconnected power systems. J Franklin Inst 2018;355(17).

[15] Khokhar B, Dahiya S, Parmar KPS. A novel fractional order proportional integral derivative plus second-order derivative controller for load frequency control. Int J Sustain Energy 2021;40(3):235–52.

[16] Latif A, Hussain SM, Das DC, Ustun TS. State-of-the-art of controllers and soft computing techniques for regulated load frequency management of single/multi-area traditional and renewable energy based power systems. Appl Energy 2020; 266.

[17] Peddakapu K, Mohamed MR, Srinivasarao P, Leung PK. Simultaneous controllers for stabilizing the frequency changes in deregulated power system using moth flame optimization. Sustain Energy Technol Assess 2022;51.

[18] Singh K. Load frequency regulation by de-loaded tidal turbine power plant units using fractional fuzzy based PID droop controller. Appl Soft Comput 2020;92: 106338.

[19] Singh K, Amir M, Ahmad F, Refaat SS. Enhancement of frequency control for stand-alone multi-microgrids. IEEE Access 2021;9:79128–42.

[20] Tasnin W, Saikia LC. Deregulated AGC of multi-area system incorporating dish-Stirling solar thermal and geothermal power plants using fractional order cascade controller. Int J Electr Power Energy Syst 2018;101:60–74.

[21] Arya Y. Effect of electric vehicles on load frequency control in interconnected thermal and hydrothermal power systems utilising CF-FOIDF controller. IET Gener Transm Distrib 2020.

[22] Oshnoei S, Aghamohammadi M, Oshnoei S, Oshnoei A, Mohammadi-Ivatloo B. Provision of frequency stability of an islanded microgrid using a novel virtual inertia control and a fractional order cascade controller. Energies 2021;14(14).

[23] Oshnoei S, Oshnoei A, Mosallanejad A, Haghjoo F. Contribution of GCSC to regulate the frequency in multi-area power systems considering time delays: a new control outline based on fractional order controllers. Int J Electr Power Energy Syst 2020;123.

[24] Latif A, Hussain SM, Das DC, Ustun TS. Design and Implementation of maiden dual-level controller for ameliorating frequency control in a hybrid microgrid. Energies 2021;14(9).

[25] Latif A, Hussain SM, Das DC, Ustun TS. Optimization of two-stage IPD-(1+ I) controllers for frequency regulation of sustainable energy based hybrid microgrid network. Electronics 2021;10(8).

[26] Zaheeruddin, Singh K. Intelligent fractional-order-based centralized frequency controller for microgrid. IETE J Res 2020:1–15.

[27] Arya Y, Kumar N. Design and analysis of BFOA-optimized fuzzy PI/PID controller for AGC of multi-area traditional/restructured electrical power systems. Soft Comput 2017;21(21):6435–52.

[28] Arya Y. Effect of energy storage systems on automatic generation control of interconnected traditional and restructured energy systems. Int J Energy Res 2019; 43(12):6475–93.

[29] Latif A, Paul M, Das DC, Hussain SM, Ustun TS. Price based demand response for optimal frequency stabilization in ORC solar thermal based isolated hybrid microgrid under salp swarm technique. Electronics 2020;9(12).

[30] Singh K, Amir M, Ahmad F, Khan MA. An integral tilt derivative control strategy for frequency control in multimicrogrid system. IEEE Syst J 2020;15(1):1477–88.

[31] Sulaiman MH, Mustaffa Z, Saari MM, Daniyal H. Barnacles mating optimizer: a new bio-inspired algorithm for solving engineering optimization problems. Eng Appl Artif Intell 2020;87:103330.

[32] Khokhar B, Dahiya S, Singh Parmar KP. A robust Cascade controller for load frequency control of a standalone microgrid incorporating electric vehicles. Electr Power Components Syst 2020;48(6–7):711–26.

[33] Yousri D, Babu TS, Fathy A. Recent methodology based Harris Hawks optimizer for designing load frequency control incorporated in multi-interconnected renewable energy plants. Sustain Energy, Grids Networks 2020:100352.

[34] Yang Z, Liu Q, Zhang L, Dai J, Razmjooy N. Model parameter estimation of the PEMFCs using improved barnacles mating optimization algorithm. Energy 2020; 212:118738.